

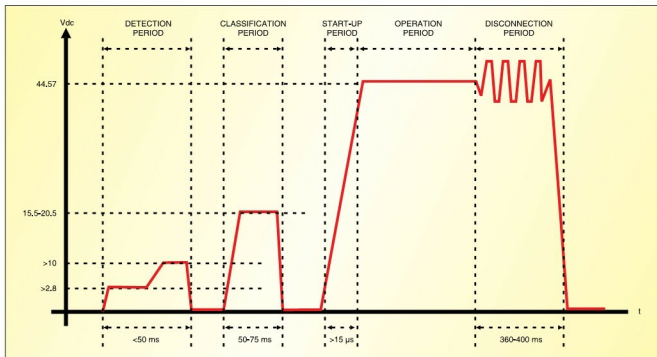


Fig. 3: PoE power-on process

stations, notebooks, video phones and motorised cameras.

Benefits of PoE

The benefits of the PoE technology are more than just eliminating the power cords. These can include cost-effectiveness, simple deployment, mobility, flexibility, reliability and enhanced management.

Cost-effectiveness. Both time and money could be saved. Without the traditional electrical circuit, not only the costs for creating and maintaining the power infrastructure such as cords, outlets and conduits but also the costs for expensive electricians are saved. The time spent in installing the power system is also eliminated.

Simple deployment. Only one set of wires is required to connect and power the appliance. Both the manufacturers and users benefit from the PoE technology. No specific accessories to suit different countries' power systems are needed.

Mobility and flexibility. Deployment of a device is no longer restricted to a short distance from the AC outlets. Thus, systems like IP cameras and

wireless access points can be easily located in remote places, such as ceiling, for security and performance.

Reliability. Connecting your UPS to the PSE can guarantee power supply even when mains power fails.

Enhanced management. Centralised power and data management, SNMP, for example, allows the IT staff to remotely monitor and manage the access points, IP cameras or other PDs.

Industrial PoE challenges

Rather than using commercial PoE apparatus to deploy PoE in industrial environments, various kinds of challenges need to be considered. Any carelessness may result in system failure or device damage. A number of issues that arise when adopting PoE in an industrial environment are highlighted below.

Reliability. Reliability is critical to ensure that industrial applications keep running. Any single-point failure may need to be recovered in a short time. Aside from the network reliability, which is discussed in Ethernet environment, the demand for stability and reliability of power

supply is rising because of power and data integration into a single infrastructure.

PSE reliability. Acting as a power source, the PSE applies power to one or more PDs. A single PSE failure may lead to a number of PDs shut down. As a result, the reliability of the PSE itself and its power input becomes critical.

Basically, the PSE should stay stable even under the most unfavourable operating conditions, such as extreme temperature, humidity, vibration and shock. A stronger PSE may further be equipped with self-protection mechanisms for undetected system failure. As regards the power input, it can be enhanced by supplying through a UPS system or other means to avoid sudden disconnection.

PD reliability. The design of a PD may get complex to sustain harsh industrial conditions. However, a smart PSE may help to detect whether the PD is still alive. A detection of loss or failure of the PD can be reported to the IT staff by various management mechanisms such as e-mail, alarm and SNMP traps. The IT staff